

Ans to the Qno-4

(i) Non-pipelined Processor:

For a non-pipelined single-cycle processor, each instruction must complete five stages within a single clock cycle. So the instruction takes a lot of time to complete all stages.

For non-pipeline processor,

$$\begin{aligned}\text{Clock Cycle Time} &= (\text{IF} + \text{ID} + \text{EX} + \text{MEM} + \text{WB}) \\ &= (280 + 200 + 210 + 280 + 200) \text{ ps} \\ &= 1170 \text{ ps}\end{aligned}$$

Pipelined Processor:

For pipelined processor, each instruction is executed partially which improves performance and reduces clock cycle time. So we take the maximum duration taken in an instruction to accommodate all the steps per stage.

$$\text{Clock Cycle Time} = \max(\text{IF}, \text{ID}, \text{EX}, \text{MEM}, \text{WB})$$

$$= \max(280, 200, 210, 280, 200) \text{ ps}$$

$$= 280 \text{ ps}$$

(Ans)

$$\text{(ii) Total time required} = (\text{Total number of stages} - 1) + \text{Number of Instructions} + \text{Total stall cycle}$$

$$= (5 - 1) + 5 + 4$$

$$= 13 \text{ clock cycles}$$

(5)(i) ADD r_{21}, r_{25}, r_{25}

LD $r_{21}, 24(r_{25})$

SD $r_{21}, 24(r_{25})$

C1	C2	C3	C4	C5	C6	C7	C8	C9	C10
IF	ID	EX	MEM	WB					
	IF	ID	EX	MEM	WB				
		IF	ID	EX	MEM	WB			

(ii) Total clock cycle = 7

Ans. (b) is correct